Reward Shaping Is Noise Shaping

and a law for the accuracy–diversity frontier of RLVR

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The question

- RL with verifiable rewards: sample answers, a checker grades them, update $g = \sum_i A_i \nabla \log \pi(y_i)$.
- Many papers change the advantage A_i (reward right only, penalize wrong only, baselines, pass@k weights, entropy bonuses) and report more **diversity** (pass@k).
- They compare methods **at the same training step**. A slower method is less sharpened there, so it *looks* more diverse.
- Our question: is the diversity real? The fair test is **pass@k at matched pass@1**.

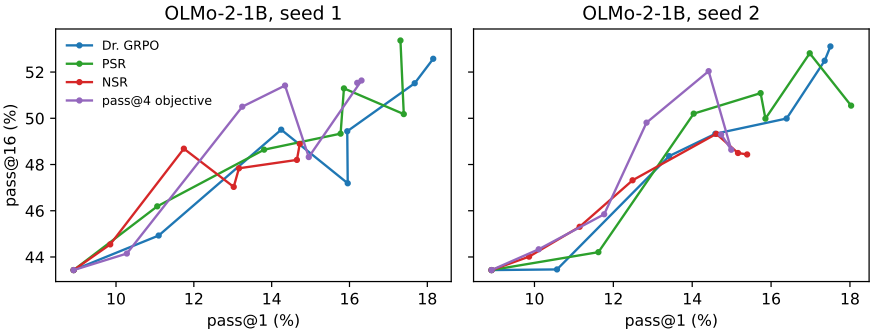
Theory: same direction, different noise

For any rule that scores answers only by correctness (p: a question's success rate)

$$\mathbb{E}g = c(p)\,\nabla p, \qquad \mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p)$$

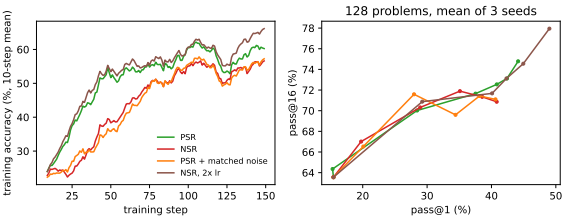
- A rule can only reweight questions and change the noise; wrong answers carry most of it (2–3×).
- Noise sets the **speed** along one accuracy–diversity curve, not the curve.
- At matched pass@1, pass@$k = \mathbb{E}_x[1 - (1 - p_x)^k]$: diversity is how evenly success is spread across questions.

Diagnosis 1: one curve, two model families



OLMo-2-1B: NSR minus PSR at matched accuracy -0.30, 90% $[-1.28, +0.77]$: equivalent within ± 2. Same on Qwen for PSR, NSR, five baselines, the NSR paper's code, a pass@4 objective.

Diagnosis 2: the noise is the cause



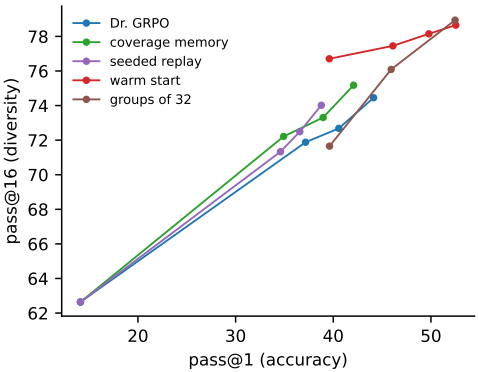
- Add zero-mean noise of NSR's size to PSR.
- **PSR + noise is as slow as NSR**: $\phi = -0.11$ $[-0.22, -0.05]$ (NSR $= 0$, PSR $= 1$).
- NSR at $2\times$ learning rate recovers.
- Not temperature either: a late checkpoint sampled hotter does not beat the curve (-1.42).

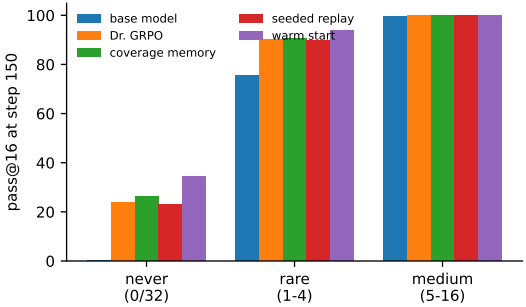
Trying to move the curve

Method	pass@16 at matched accuracy	
Reward shaping, entropy, unlikeliness, pass@k objective	≈ 0	finished
Replay of own successes (coverage-weighted)	$+1.19$ $[+0.54, +1.79]$	finished
Replay seeded with base-model successes	$+1.02$; 5.4 points less accurate	finished
Annealed replay; MaxRL + group completion (pilot)	$+1.3$; $+1.1$; 6.5 less accurate	interim
Group completion; prefix-guided rollouts (pilot)	–	evaluating
Warm start (fine-tune on base successes, then RL)	**beyond Dr. GRPO's range**	
Groups of 32 ($4\times$ samples)	**beyond Dr. GRPO's range**	2 of 3 seeds

Inside RL, everything sits near $+1$, below our $+2$ bar.

Where the gains go





Warm start and groups of 32 end at the same point (pass@1 52.5, against 44.1 for Dr. GRPO). The warm start's pass@16 gain sits on problems the base never solves (24 $\rightarrow$ 34). At 1,024 tokens all numbers rise (Dr. GRPO pass@1 57.7 vs 44.6): the 512-token budget matters.

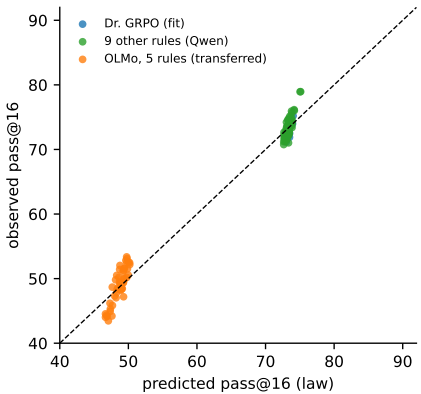
New: a law for the frontier

Per-question log-odds learning from base counts

$$\theta_x(0) = \log \frac{c_x + a}{n - c_x + b}, \qquad \dot{\theta}_x = p_x(1 - p_x) + \tau \, \overline{p(1-p)}$$

- Inputs: the base model's per-question success counts; each checkpoint's pass@1 (sets progress t).
- $\tau = -0.20$ fitted once on Dr. GRPO, then **frozen** (pre-registered).
- Predicts the whole distribution: pass@k for all k, difficulty strata, per-question counts.
- Diversity collapse = **rich get richer**: a uniform improvement would put never-solved problems at 70–80 pass@16; observed 24.

The law, tested prospectively



Test set	RMSE pass@16	verdict
9 other rules, Qwen (56)	**1.21**	supported
OLMo, τ from Qwen (50)	**1.90**	supported
1,024 tokens (4)	**0.85**	supported
Warm start (9)	3.56	falsified
450 steps	running	

Alternatives: uniform shift 10–21, affine 1.3–12.6; the law wins in per-question likelihood in 117/119 checkpoints.

What the law says

- From a base model, the frontier is set by **base difficulty** and near-independent per-question learning; reward rules only change how far along it a run gets.
- That is why reward shaping cannot move it. Only two levers can: the **starting distribution** (information on rare questions) and **transfer** between questions.
- **The warm start breaks the law**: afterwards RL lifts never-solved problems 12 points less than predicted (fitted $\tau \approx -0.7$): fine-tuning changes the dynamics.
- Against the law's extended frontier the warm start sits $+3.4$ above and groups of 32 reaches $+3.9$; the 450-step run decides whether that is real.

Position in the 2026 literature

Ours, unpublished

- noise sets speed along one curve
- NSR on PSR's curve, two families
- the noise-injection causal test
- matched-accuracy protocol
- temperature does not beat the curve
- a prospective frontier law

Published, to cite or answer

- same direction, per-question weight (Davis & Recht; Heckel et al.)
- base-seeded replay (TAR); coverage weight (BBG)
- rescue of all-wrong groups (LSPO, LUFFY)
- transfer curricula (TAC, GradAlign)
- "RLVR is a Kernel": strongest challenge to the law

No 2026 method paper compares at matched accuracy.

Where we stand

Stage	Independent review	Why
Diagnosis only	5/10	solid, but the theory is partly prior work
+ law passes prospective tests	**7/10**	a new, predictive result
+ method that improves the frontier across scales	8/10	diagnosis → law → method

- **Running:** 450-step Dr. GRPO (law beyond its fit); deep base sampling (sign of τ); TAR, BBG, MaxRL, W-REINFORCE at matched accuracy; pilot evaluations; 1,024-token and transfer checks.
- **Method candidate from the law:** spend demonstration acquisition where the law predicts stalls, against random allocation and the warm start at equal compute.